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Atlantic Technological University

Comparative Evaluation of Existing Phishing and Malicious Email Detection Frameworks (2024–2026)

Reproducing an interpretable ML approach, a BERT hybrid, and typed entity-relation detection

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Abstract

Phishing emails remain a persistent threat because malicious messages may resemble routine communications and evade simple keyword rules. This dissertation evaluates three existing methodological families from 2024, 2025 and 2026 rather than proposing a new detection framework. The 2024 approach combines term frequency–inverse document frequency (TF-IDF), conventional classifiers and local explainability. The 2025 approach couples BERT embeddings with multi-head attention, a one-dimensional convolutional neural network and a bidirectional gated recurrent unit. The 2026 PhishSigma++ approach models typed entity relations from raw RFC822 email and selects sparse features by particle swarm optimisation.

A structured literature comparison is combined with three controlled reimplementations whose outputs were supplied as executed Colab reports. In the 2024 experiment, linear SVC achieved an F1-score of 0.9914 on 16,414 test emails. In the 2025 controlled comparison, plain BERT achieved F1 = 0.9710, exceeding the fixed hybrid (F1 = 0.9640) on the same 1,864-item test set. In the 2026 reproduction, typed-relation detection achieved F1 = 0.9999, whereas its TF-IDF/Naive Bayes comparator achieved F1 = 0.8833; however, strong dataset-source confounding makes the near-perfect result unsuitable as proof of real-world generalisation. The three original tasks, corpora and protocols differ, preventing a valid cross-paper performance ranking. For a lightweight and explainable email-text application, the 2024 TF-IDF/linear SVC approach is the most defensible provisional choice; standard BERT is preferable to the fixed hybrid within the 2025 paired test, while the 2026 approach remains a promising but insufficiently validated research candidate. A common source-disjoint benchmark is required before declaring a universal winner.

Keywords: phishing email detection; malicious email; TF-IDF; support vector machine; BERT; explainable AI; typed entity relations; robustness; reproducibility.

Acknowledgments

I would like to thank my supervisor, Dr Owen Foley, for his guidance in defining the project scope, selecting a comparative evaluation approach and emphasising the use of datasets and practical evidence. I also acknowledge Atlantic Technological University and the authors and maintainers of the public research papers, datasets and open-source software used in this study.

Abbreviations

Abbreviation	Meaning
AUC	Area under the curve
BERT	Bidirectional Encoder Representations from Transformers
BiGRU	Bidirectional gated recurrent unit
CNN	Convolutional neural network
CV	Cross-validation
F1	Harmonic mean of precision and recall
FN / FP	False negative / false positive
LIME	Local Interpretable Model-Agnostic Explanations
MGO	Mountain Gazelle Optimizer
MNB	Multinomial Naive Bayes
NLP	Natural language processing
PSO	Particle swarm optimisation
RFC822	Internet email message format (legacy designation)
SVC / SVM	Support vector classifier / machine
TF-IDF	Term frequency–inverse document frequency
XAI	Explainable artificial intelligence

Chapter 1

Introduction

1.1 Background and motivation

Email remains central to communication across professional, educational and personal settings. Phishing messages impersonate trusted organisations, create urgency or exploit familiar business processes to persuade recipients to reveal credentials, transfer money or follow unsafe links. A detector must therefore distinguish suspicious messages from genuine account notifications, invoices and other ordinary correspondence. As language models make deceptive emails easier to produce and vary, surface-level words alone may not be enough for sustained protection [1, 2, 3].

Existing academic work explores several complementary designs. Classical machine learning converts text into statistical word features and learns a decision boundary. Transformer models encode words in context and can be extended by attention, convolutional and recurrent layers. Entity-relation methods instead focus on relationships among sender addresses, domains, recipients, links, subject text and socially engineered cues. These methodological differences provide the rationale for the present comparison.

1.2 Research problem

Published studies often report high accuracy or F1-score but use different email corpora, labels, feature availability, splits, compute budgets and attack settings. These results are informative within their own experiments but cannot establish which system is best without considering the evaluation design. A further problem is reproducibility: the exact source code, preprocessing, sample assignments or optimisation settings may not be available. The central problem is therefore how to evaluate existing detection approaches fairly and critically using available code, public datasets and transparent reporting.

1.3 Aim and research question

Aim: To critically evaluate three existing email detection methodologies from 2024, 2025 and 2026 using a structured literature review and controlled reimplementation, examining classification effectiveness, interpretability, robustness, cost and methodological limitations.

Primary research question: How do representative interpretable machine-learning, contextual hybrid deep-learning and typed entity-relation methods compare within their respective controlled experiments, and what can be concluded about practical suitability given their different datasets and protocols?

1.4 Objectives

Table 1.1 lists the research objectives and the evidence used to judge whether each was met.

Objective	Success evidence
O1. Review the selected publications	A traceable comparison of algorithms, corpora, methods and reported results.
O2. Recreate core methodology	Executable implementations and documented departures from each original study.
O3. Evaluate discrimination	Test accuracy, positive-class precision, recall, F1 and confusion matrices.
O4. Assess interpretability and robustness	LIME explanations for 2024; entity evidence and Good Word test for 2026.
O5. Critically assess validity	Explicit analysis of duplicates, source-label coupling, split fairness and task-definition mismatch.
O6. Draw a cautious recommendation	A use-case-aware synthesis rather than an unsupported cross-dataset ranking.

Table 1.1: Research objectives and observable evidence

1.5 Scope and exclusions

The scope is detection of phishing or broader malicious content in email messages. The project compares existing systems and does not claim development of a novel phishing classifier. It does not evaluate web-page-only detection, URL-only datasets or live email account integration. The 2026 experiment intentionally

adopts the broader "malicious" positive class used in its source implementation, including some spam; this makes its task different from a strictly phishing-versus-legitimate evaluation. Practical evaluation uses offline public data, not deployment in a production gateway.

1.6 Contribution and thesis organisation

The contribution is a documented comparative reproduction with evidence of how methodological choices affect apparently strong results. It includes a transparent 2025 plain-BERT-versus-hybrid test, a 2024 three-classifier comparison with local explanations and a 2026 typed-relation experiment with a robustness curve. All project code and notebooks are available in the accompanying GitHub repository [4]. The thesis follows the ATU template: Chapter 2 explains research methodology; Chapter 3 reviews the literature; Chapter 4 describes implementation; Chapter 5 reports the experiments and their limitations; and Chapter 6 concludes. Appendices provide artefact traceability and additional published results.

Chapter 2

Research Methodology

2.1 Research design

This project follows an empirical, quantitative, comparative case-study design supported by a structured reading of three selected publications. Each paper forms an implementation case. The original results are first recorded as literature evidence. Student code then operationalises the main pipeline on accessible data, saves model predictions and computes shared binary metrics. The work is a transparent methodological reproduction rather than a claim that the original software environment, optimisation or exact data split has been recreated. The overall workflow is shown in Figure 2.1.

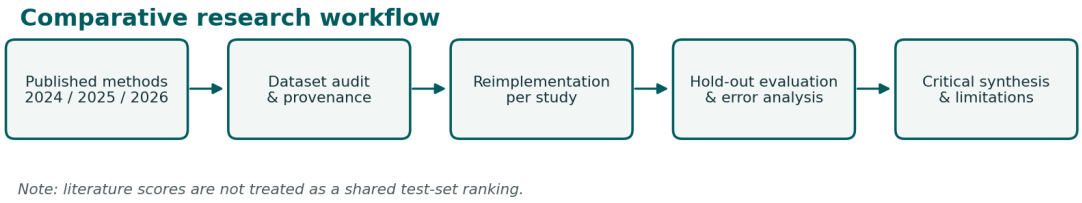


Figure 2.1: Comparative research workflow.

2.2 Selection of papers

The three studies were chosen to sample substantively different feature representations while staying within the supervisor-approved scope of email phishing or malicious-content detection: (i) an interpretable web-oriented text-classification framework from 2024 [1], (ii) a hybrid BERT architecture with adaptive optimisa-

tion from 2025 [2], and (iii) the typed entity-relation PhishSigma++ preprint from 2026 [3]. Selection was purposive, not a systematic review of every publication.

2.3 Dataset strategy and label alignment

Table 2.1 summarises the corpora and task definitions used in each case.

Case	Dataset/corpora	Task definition	Source / note
2024	Kaggle merged phishing-email corpus; 82,486 initial rows	Dataset label 0/1; positive class includes spam/phishing-labelled source mixtures	naserabdullahalam/ phishing-email-dataset [5]
2025	Kaggle Phishing Emails; 18,650 initial rows	Safe Email (0) vs Phishing Email (1)	subhajournal/ phishingemails [6]
2026	Enron; Phishing Pot; Nazario; SpamAssassin; 28,386 assembled rows	Benign (0) vs malicious/spam (1)	Four public raw-email sources; source imbalance [7, 8, 9, 10]

Table 2.1: Dataset inventory and task definitions

For each case, the experimental test partition was held out during classifier training. The 2024 run used an 80/20 stratified split after cleaning; the 2025 run used an 80/10/10 stratified train/validation/test partition and fixed model settings; the 2026 run used a 75/25 stratified split and inner cross-validation for PSO fitness. Since these are distinct experimental protocols and corpora, results may be compared directly only within a case where samples, labels and metric definitions are shared.

2.4 Preprocessing, reproducibility and information leakage

Data processing was recorded before model fitting. The 2024 notebook reported 408 exact duplicate source rows and removed 417 duplicate entries after text cleaning; the two counts are not contradictory because normalisation may collapse additional strings. The 2025 notebook detected 1,112 repeated email texts yet set `REMOVE_DUPLICATES=False`, so duplicates could appear across training and test partitions. This does not prove leakage occurred in every repeated record, but

creates an unresolved risk that must be checked using text hashes. The 2026 run assembled strongly source-dependent benign and malicious corpora, which can allow a model to distinguish dataset provenance rather than phishing intent. These risks affect the strength of the conclusions in Chapter 5.

2.5 Evaluation measures

All reported binary metrics treat the positive class as phishing or malicious according to the case-specific target. Let TP denote correctly classified positive emails, TN correctly classified negative emails, FP benign emails incorrectly flagged as positive and FN positive emails incorrectly classified as benign.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2.1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2.2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2.3}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2.4}$$

Accuracy measures overall correct classification; precision measures the proportion of positive alarms that are correct; recall measures the proportion of true attacks detected; and F1 balances precision and recall. False negatives are particularly important when missed phishing can produce harm. Confusion matrices and paired McNemar comparisons supplement aggregate metrics when two models predict the same test examples. A McNemar result does not by itself establish superiority outside the tested corpus.

2.6 Explainability, adversarial testing and computational costs

Interpretability was inspected through LIME [11] for the 2024 linear SVC and through the retained typed-relation mask for the 2026 approach. The 2026 experiment also used a non-adaptive Good Word perturbation, inserting benign text into malicious test emails at five increasing injection rates while intending to preserve header and URL signals. For 2025, training time was recorded for both models. These secondary measures are informative but heterogeneous; direct numerical comparison of explainability or robustness across all three cases was not performed.

2.7 Ethics and limitations of the research method

Only public historical datasets and isolated offline experiments were used. No emails were sent to recipients and no deceptive content was deployed. Public corpora can contain real names, addresses and personal communication; therefore, identifiable sample texts should be minimised in dissemination and replaced by redacted excerpts. The results are constrained by corpus age, possible mislabelled spam, repeated records, incomplete experimental specification and differing feature access. In particular, the extraordinarily high 2026 held-out accuracy requires source-disjoint follow-up evaluation before operational claims are made.

Chapter 3

Literature Review

3.1 Overview

The literature demonstrates a methodological shift from lexical classification to contextual representation and structured evidence. The three reviewed papers differ in target labels, deployment assumptions and attack models. This chapter distinguishes their published findings from results produced by the experiments in Chapter 5, and does not treat scores obtained on different corpora as directly comparable.

3.2 Al-Subaiey et al. (2024): interpretable machine learning

Al-Subaiey and colleagues propose a web-oriented phishing-email classification system that merges multiple public corpora and compares term-frequency features and classifiers [1]. After normalising and combining email fields, TF-IDF or Word2Vec features are provided to Multinomial Naive Bayes, Random Forest and SVM. The strongest paper configuration uses a linear SVM, followed by LIME to show the contributions of individual words and a web interface to support real-time user interaction. The journal article reports approximately 0.99 F1 for the best configuration. This design is comparatively transparent and deployable, but its dependency on corpus composition and lexical features requires further scrutiny.

In an email-security context, a high TF-IDF performance score may partly reflect repeated phrases or campaign artefacts in a merged dataset. LIME creates local approximations of model predictions; it does not provide a causal justification or guarantee that highlighted words represent a phishing tactic. The present work tests the principal TF-IDF classifiers and generates one local explanation, but does

not evaluate the original Flask user interface.

3.3 Hosseinzadeh et al. (2025): contextual hybrid learning

Hosseinzadeh et al. describe a framework that combines contextual embeddings from BERT [12] with multi-head attention, convolutional feature extraction, recurrent sequence modelling and Mountain Gazelle Optimizer (MGO) hyperparameter search [2]. The stated purpose is to detect context-dependent phishing patterns while balancing performance with regularisation. The paper reports headline abstract values of 96.8% accuracy, 97.2% precision, 95.4% recall and 96.3% F1. Its detailed tables report a BCG-MHeadAttention-MGO result of 0.9722 accuracy, 0.9426 precision, 0.9871 recall and 0.9643 F1. Both reported sets are acknowledged because they are not identical.

In the implementation, a plain fine-tuned BERT classifier is introduced as an explicit baseline and compared against a fixed hybrid using the same tokenizer and sample split. The architecture is a disclosed interpretation, not an exact reproduction of unspecified layers or optimisation. MGO is left disabled in the supplied executed notebook; therefore, the hybrid result in Chapter 5 must not be described as an MGO-optimised outcome.

3.4 Shang et al. (2026): typed email entity relations

PhishSigma++ approaches email detection through typed relationships among sender identities, domains, subjects, URLs, recipient features, message headers and socially engineered keywords [3]. Its 40 entity types are compared using five relation functions and collapsed into a 40×40 numeric feature representation. PSO [13] selects a sparse relational mask, after which a linear SVM performs classification. The authors report F1 = 0.9675 on 29,142 clean messages and F1 = 0.9579 under a strong Good Word padding setting. The approach is intended to preserve relational evidence even if malicious body text is diluted with harmless language.

PhishSigma++ is an arXiv preprint, not demonstrated in the sources as a peer-reviewed journal article. It handles broader malicious mail and has a different input requirement from text-only approaches: full RFC822 records with meaningful headers are needed. The reimplementation has deliberately simplified lexical dictionaries and source inclusion; it is valuable as a proof of methodology

but should not be equated with the original paper implementation.

3.5 Literature comparison

Table 3.1 summarises the three approaches.

Year	Representation	Classifier / selection	Interpretability	Key limitation
2024	TF-IDF; Word2Vec comparator	SVM, MNB, RF	LIME words	Vocabulary/corpus shift
2025	BERT context + atten-tion/CNN/BiGRU	Dense classifier; MGO tuning	Limited native model explanation	Compute and implementation ambiguity
2026	40 typed entities × 5 relations	PSO mask + linear SVM	Relation-type evidence	RFC822 need; source/label confounding

Table 3.1: Comparative literature summary

3.6 Additional baselines described by the 2025 study

Published model	Accuracy	Precision	Recall	F1
CNN	0.8937	0.8376	0.8953	0.8655
LSTM	0.9106	0.8616	0.9041	0.8823
BiLSTM	0.9224	0.8789	0.9252	0.9014
GRU	0.9258	0.8839	0.9279	0.9054
CLSTM	0.9106	0.8616	0.9166	0.8882
RCNN	0.8988	0.8449	0.8983	0.8708
ServeNet	0.9325	0.8938	0.9361	0.9145
CARL-Net	0.9376	0.9015	0.939	0.9199
BCG-MHeadAttention-MGO	0.9722	0.9426	0.9871	0.9643

Table 3.2: Published 2025 model comparison (study Table 9) [2]

The published comparator table (Table 3.2) does not supply our own results for all of these architectures. It is presented here only to describe what the existing 2025 study evaluated. The 2025 run compares plain BERT with the fixed hybrid, not CNN/LSTM/ServeNet/CARL-Net as independently retrained models. The study also reports GloVe/BERT embedding, dropout/L2 ablation and multiple optimiser comparisons; these were reviewed rather than fully reproduced.

3.7 Research gap and analytical framework

The studies show strong performance within selected evaluation conditions. A key research gap is separating held-out performance from generalisation to new organisations, campaigns and corpora. The present work therefore distinguishes literature values, reproduced within-study outcomes and provisional inferences requiring further validation. It avoids treating heterogeneous scores as interchangeable.

Chapter 4

System Design and Implementation

4.1 Overview of reproducible experimental artefacts

Each notebook was implemented in Google Colab. The 2024 pipeline primarily uses scikit-learn [14]; the 2025 pipeline uses PyTorch and Hugging Face Transformers [15] with a Tesla T4 GPU in the captured execution; and the 2026 pipeline uses Python email parsing, NumPy, a PSO feature search and scikit-learn. The version-controlled notebooks and project code are publicly available in the project GitHub repository [4]: `https://github.com/Azeem74812/Msc-Thesis-Project.git`. Source CSV checksums and full machine logs should also be archived before the final institutional submission.

4.2 2024 design

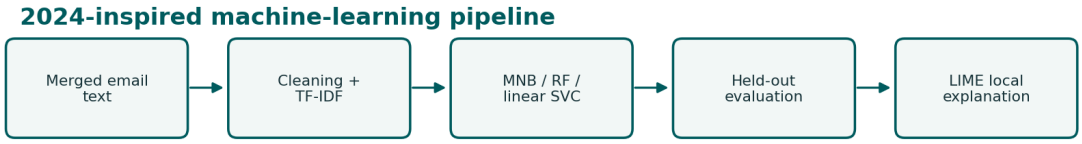


Figure 4.1: 2024-inspired machine-learning pipeline.

The notebook (Figure 4.1) loads the merged Kaggle file, confirms the `text_combined` and `label` columns, maps the binary labels, lowercases and normalises text, re-

places URLs with a generic marker, removes punctuation and drops duplicate cleaned strings. A stratified 80/20 train/test split is established using seed 42. Three classification pipelines fit TF-IDF on training text only and evaluate on the identical test data: Multinomial Naive Bayes, a 100-tree Random Forest, and a probability-enabled linear-kernel SVC. LIME samples perturbations of a selected test email and presents approximate local feature weights. A paired McNemar test contrasts SVC and Naive Bayes.

4.3 2025 design

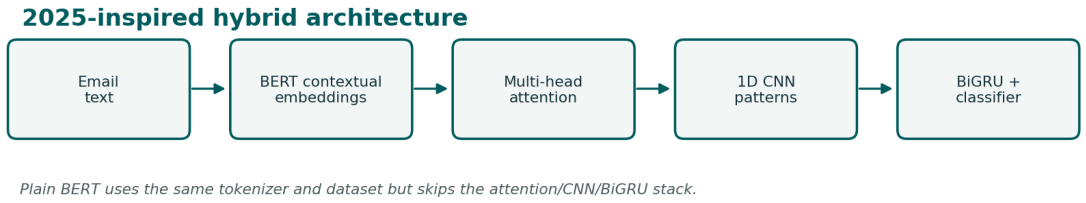


Figure 4.2: 2025-inspired BERT hybrid architecture.

The notebook downloads the 2025 paper-associated Kaggle dataset and creates a stratified 80/10/10 train/validation/test split. Both models use `bert-base-uncased`, 256-token truncation, three epochs, AdamW, weight decay and a consistent validation rule. Baseline BERT applies a standard sequence-classification head. The fixed hybrid (Figure 4.2) uses token-level BERT representations, batch normalisation, multi-head attention, 1D convolution, BiGRU, masked mean pooling and dense layers. The paper description of pooled BERT output is not fully consistent with sequence-level attention/CNN/GRU processing; the implementation explicitly interprets this as token-level `last_hidden_state`. Model selection is based on validation F1, while the same test set is used for the final comparison.

One relevant distinction is that the optional MGO tuning and regularisation-ablation cells were disabled in the supplied execution. The hybrid evaluated in Chapter 5 therefore reflects fixed experimental hyperparameters rather than the fully optimised architecture claimed in the source paper.

4.4 2026 design

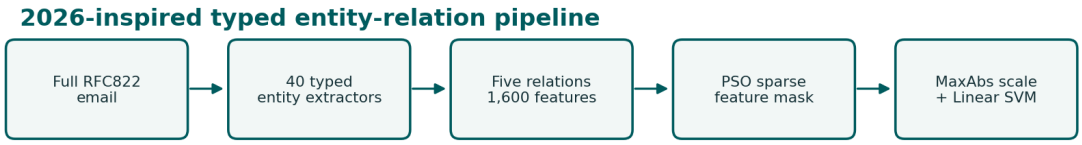


Figure 4.3: 2026-inspired typed entity-relation pipeline.

The 2026 notebook (Figure 4.3) obtains raw message records from Enron, Phishing Pot, Nazario and SpamAssassin sources and stores a `raw_message`, binary label and source column. Forty extractor types cover sender identity, routing, recipients, URL structures, attachment, subject/body content and keyword indicators. Five pairwise relation functions measure exact equality, containment, longest common substring overlap, token Jaccard overlap and length ratio. Relations are collapsed into 1,600 feature coordinates. PSO searches a sparse real-valued mask using three-fold inner training cross-validation with a sparsity penalty; the final features are MaxAbs-scaled and classified with linear SVC.

The student run uses 25 PSO iterations rather than the original paper's 50, although other reported parameters include 30 particles, inertia 0.7 and attraction weights 1.5. The code processes up to five entity occurrences per type when building relations, an explicit runtime approximation. Labels conflate spam with other malicious mail as described in the notebook. For error analysis and interpretability, the proposed top-relation evidence must be independently verified: the printed code appears to assign coefficients to a thresholded-mask index list even though sklearn `LinearSVC` coefficients may remain indexed by all 1,600 features. Consequently, the displayed evidence weights cannot be treated as audited explanations without code correction.

4.5 Design decisions, controls and artefact traceability

Case	Data split	Features	Model control	Unreproduced element
2024	80/20 stratified	40,000 TF-IDF maximum	seed 42; 100 RF trees	Original web deployment not tested
2025	80/10/10 stratified	BERT 256 tokens	3 epochs; T4 GPU; same data both models	MGO optimisation disabled
2026	75/25 stratified + 3-fold PSO inner CV	40 types; 1,600 features	30 particles; 25 iterations; 42	Original full lexicons/ source-balanced split

Table 4.1: Main student implementation settings

This architecture inventory (Table 4.1) supports the reproducibility principle: readers should be able to distinguish a paper-inspired implementation from code authored by the original researchers. In addition, common functions such as label mapping, data cleaning and confusion-matrix generation are reported because they influence the measured numbers. The associated published papers and dataset links are listed in the References and Appendix A.

Chapter 5

System Evaluation and Critical Discussion

5.1 Evaluation overview

This chapter presents numerical outputs. Unless stated otherwise, values are from the experimental runs ("our result"), not the originating publication ("paper result"). Internal comparisons on a shared held-out split support local model selection. Cross-study comparisons are descriptive only because class definitions, data sources, preprocessing and experimental controls differ.

5.2 Dataset audit and scope of evaluation

Case	Initial count	Final / assembled	Split	Notable issue
2024	82,486	82,068	65,654 train / 16,414 test	417 texts removed after cleaning
2025	18,650	18,631	14,904 / 1,863 / 1,864	1,112 duplicate original texts; no deduplication
2026	Four source corpora	28,386	21,289 train / 7,097 test	Source-label overlap; only 3 SpamAssassin records parsed

Table 5.1: Input data and observed experimental partitions

For 2024, the cleaned dataset comprised 39,233 negative and 42,835 positive examples. For 2025, the retained file included 11,322 safe and 7,309 phishing emails. For 2026, the source counts were Enron 15,000, Phishing Pot 12,270, Nazario 1,113 and SpamAssassin 3, summing to 28,386. The 2026 resulting data is markedly different from the full mixed corpus reported in the paper; the Nazario parser printed an error, and almost no SpamAssassin samples were parsed. Figure 5.8 makes the provenance issue visible.

5.3 2024 results: TF-IDF classifiers

Model	Accuracy	Precision	Recall	F1	FN	FP
Multinomial NB	0.9770	0.9831	0.9726	0.9778	235	143
Random Forest	0.9854	0.9874	0.9846	0.9860	132	108
Linear SVC	0.9910	0.9894	0.9935	0.9914	56	91

Table 5.2: Our 2024 held-out model results

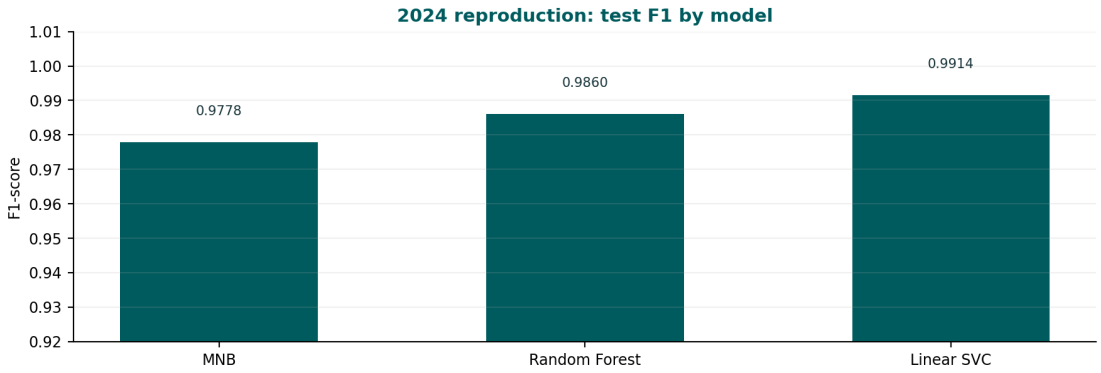


Figure 5.1: F1-scores measured in the 2024 student reproduction.

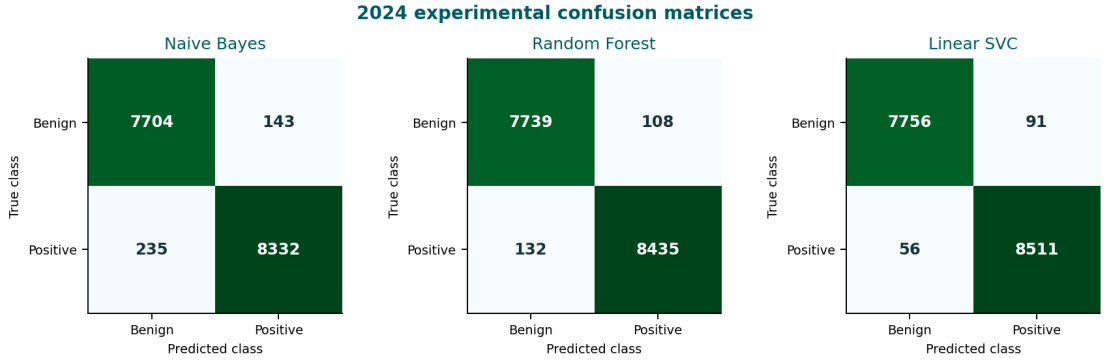


Figure 5.2: 2024 held-out confusion matrices (negative class first).

The linear SVC had the strongest F1-score (Table 5.2, Figures 5.1 and 5.2), detecting 8,511 of 8,567 positive emails while missing 56 and incorrectly flagging 91 of 7,847 negative emails. Its empirical F1 of 0.9914 is close to the approximately 0.99 published F1. Random Forest outperformed Naive Bayes but remained below SVC. The notebook reports 291 examples where Naive Bayes was wrong and SVC correct, versus 60 in the opposite direction; its exact McNemar test printed an extremely small p-value. This supports a within-dataset distinction between these classifiers, not a claim about all unseen modern phishing.

The LIME illustration provides local token-weight evidence. Some highlighted features included non-semantic fragments such as dates or adult-content tokens, which illustrates a limitation: the model may be exploiting dataset-specific lexical regularities. Because the positive class of the merged dataset includes spam/fraud material, the performance should be described as classification under that dataset's labels rather than a pure estimate of phishing-only detection.

5.4 2025 results: plain BERT versus fixed hybrid

Model	Accuracy	Precision	Recall	F1	Train min	FN / FP
Baseline BERT	0.9769	0.9587	0.9836	0.9710	16.61	12 / 31
Fixed BERT hybrid	0.9716	0.9581	0.9699	0.9640	17.66	22 / 31

Table 5.3: Our 2025 shared-test comparison

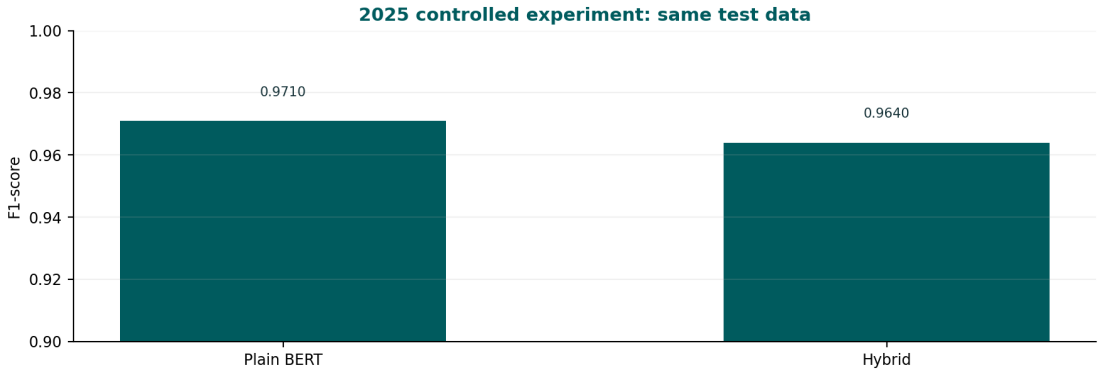


Figure 5.3: The fixed hybrid underperformed plain BERT on the supplied shared test partition.

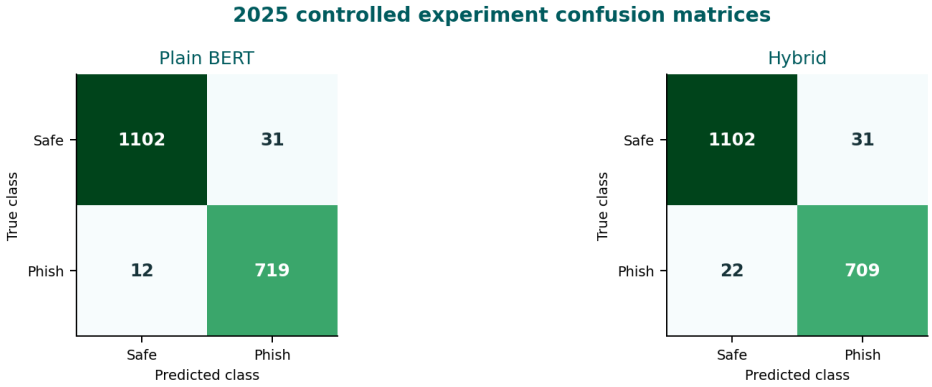


Figure 5.4: Confusion matrices for the 2025 controlled run.

Contrary to the initial expectation that adding attention, CNN and BiGRU would necessarily improve the classifier, the fixed hybrid produced an F1-score 0.0070 lower than plain BERT (about 0.70 percentage points; Table 5.3). Accuracy was 0.536 percentage points lower, and the hybrid generated 10 additional false negatives while leaving false positives unchanged at 31 (Figure 5.4). Training was about 1.05 minutes longer. The notebook's McNemar count records seven cases improved by the hybrid and 17 made worse; the paired data therefore does not support a claim that the hybrid was superior in this run.

This comparison isolates the additional architecture within the fixed settings, but it does not evaluate an MGO-selected optimum. The 2025 detailed paper's hybrid F1 = 0.9643 and the fixed-hybrid F1 = 0.9640 are numerically close, but this resemblance does not demonstrate identical replication because exact test membership, tokenizer truncation, paper checkpoint and hyperparameters are not

matched. The paper also differs between its abstract summary and detailed tables. The central defensible result is that a plain BERT baseline was stronger under the uploaded controlled experiment.

5.5 2026 results: entity relation detection

Model	Accuracy	Precision	Recall	F1	FN / FP
TF-IDF + Multinomial NB	0.900944	0.993650	0.794979	0.883281	—
Typed relations + PSO + linear SVC	0.999859	1.000000	0.999701	0.999851	1 / 0

Table 5.4: Our 2026 held-out results

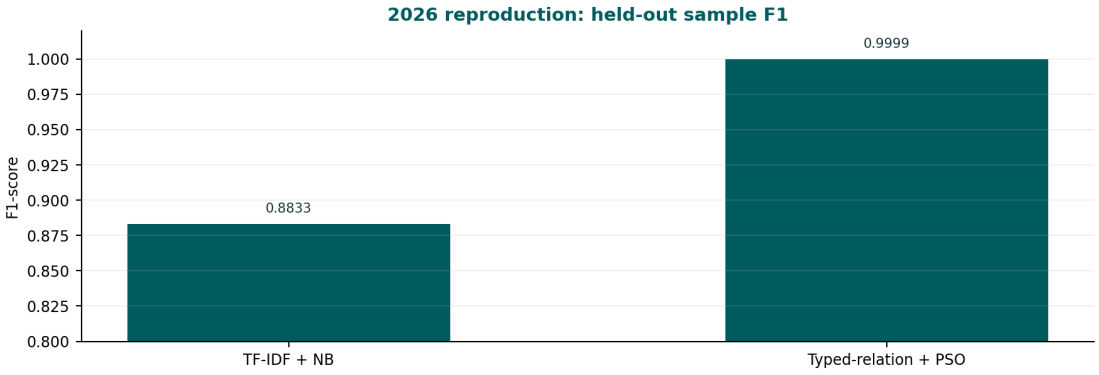


Figure 5.5: The near-perfect within-corpus 2026 result compared with its lexical baseline.

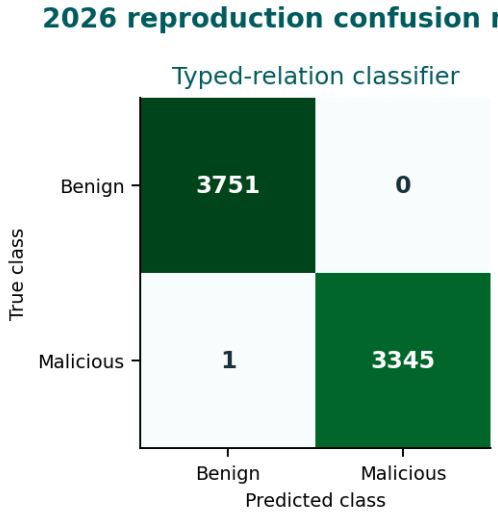


Figure 5.6: The 2026 typed-relation confusion matrix (3,751 TN, zero FP, one FN, 3,345 TP).

On the sampled 7,097-item held-out partition, the student typed-relation implementation made just one error and yielded F1 = 0.999851 (Table 5.4). By comparison, the TF-IDF + Naive Bayes token baseline gave F1 = 0.883281. The source paper reports a materially lower F1 of 0.9675 on its own 29,142-message corpus. The 2026 result must not be presented as evidence of superior generalisation to the published approach: the assembled data and source coverage differ substantially, and nearly all negative messages came from Enron while nearly all positives came from Phishing Pot or Nazario.

A further warning is that the "entity counts only", "all interaction features" and "PSO-pruned" ablation rows printed exactly the same metrics (accuracy 0.999859; F1 0.999851). This weakens any empirical claim that the learned pairwise relations or PSO selection caused the accuracy gain. A source-held-out evaluation and ablation sanity check are therefore required before attributing the near-perfect outcome to relational invariants.

5.6 Robustness to non-adaptive Good Word padding

ρ	Our relations	Our TF-IDF+NB	Paper Phish-Sigma++	Paper SpamBayes
0.0	0.999701	0.879560	0.9675	0.9672
0.2	0.999701	0.869770	0.9642	0.8080
0.4	0.999851	0.861929	0.9642	0.3299
0.6	0.999701	0.836533	0.9612	0.0633
0.8	0.999701	0.767229	0.9579	0.0243

Table 5.5: Good Word F1 results: student run and published data

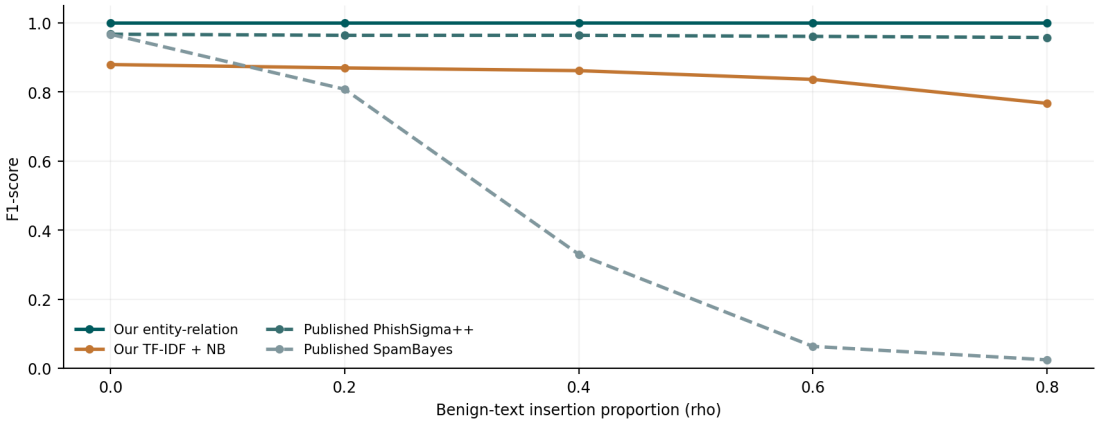


Figure 5.7: F1 against increasing benign-body padding; solid lines are the supplied student run and dashed lines are literature results.

The Good Word experiment perturbs malicious test bodies while intending to retain structural headers. The entity-relation classifier remained near 0.9997 F1 across all five injection proportions (Table 5.5, Figure 5.7), whereas its text-only Naive Bayes comparator declined from approximately 0.88 to 0.77. The publication reports a reduction for PhishSigma++ from 0.9675 to 0.9579, versus a much stronger collapse for SpamBayes. However, the curves should not be interpreted as matching the paper's exact perturbation or evaluation, particularly given class/source confounding and differences in the benign-text sampling and record conversion.

5.7 Relationship between published and experimental metrics

Case / model	Published F1	My F1	Interpretation
2024 linear SVC	≈0.9900	0.991438	Close within similar merged-dataset approach; preprocessing may differ
2025 2025-style hybrid	0.9643	0.963970	Fixed, non-MGO hybrid; scores not replication evidence
2025 plain BERT comparator	Not directly given as a standalone model	0.970966	New within-study reference baseline
2026 typed relations	0.9675	0.999851	Not directly comparable; dataset/source confounding likely

Table 5.6: Key literature results versus available student outcomes

The papers report different original tasks and distributions. Moreover, the 2024 outcome measures a merged text dataset; the 2025 outcome measures an 18,650-row text dataset; and the 2026 outcome measures an assembled raw-header dataset dominated by corpus-specific labels. Figure 5.8 explains why directly ranking the 2026 0.9999 above 2024 0.9914 and 2025 0.9710 would be methodologically unsound.

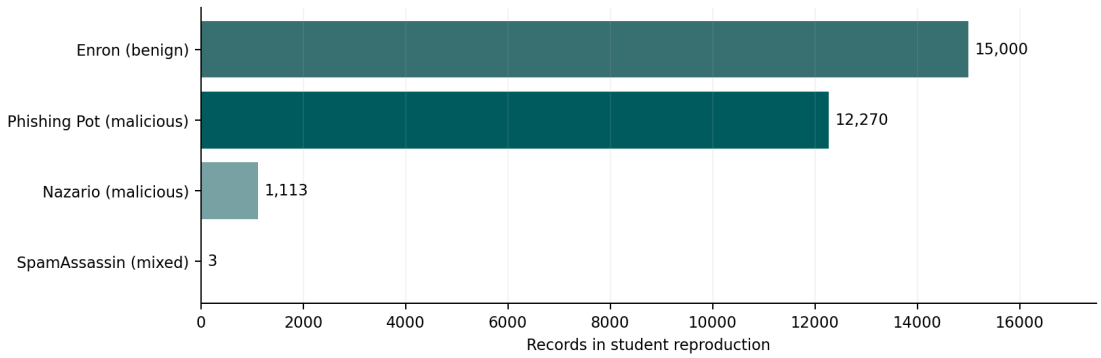


Figure 5.8: Corpus-level class provenance in the student 2026 reproduction.

5.8 Threats to validity and interpretation

Threat	Evidence	Necessary mitigation
2024 mixed positive labels	Merged corpus includes spam/fraud alongside phishing	Report target consistently; construct curated phishing-only test subset
2025 duplicate overlap	1,112 repeated text values before the split; deduplication disabled	Hash before split and repeat with group-disjoint data
2025 optimiser not used	Executed MGO and ablation switches are False	Do not call fixed hybrid "MGO optimised"; rerun separately if claiming it
2026 source/label confounding	Enron ≈ negative; Phishing Pot/Nazario ≈ positive; 3 SpamAssassin	Source-held-out and source-balanced evaluation
2026 ablations all identical	Counts, dense, PSO scores match to printed precision	Investigate redundant signals and check feature contributions
2026 implementation differences	25 versus paper 50 PSO iterations; heuristic lexicons	Describe transparent approximation and verify against official methods
Different corpora across years	No single common test set across all methods	Avoid a cross-paper winner claim; create a common raw-email benchmark

Table 5.7: Risks affecting the validity of the observed results

Additional checks include reporting confidence intervals for test F1, repeating train/test seeds, examining label inconsistencies and duplicate message hashes, and measuring inference time on identical hardware.

5.9 Practical interpretation and selection guidance

For a low-resource text-only deployment where interpretability matters, TF-IDF + linear SVC is attractive: it was fast relative to transformer-based models and supported LIME explanations in the 2024 reproduction. For richer contextual modelling, plain BERT is a strong reference: it exceeded the fixed hybrid on the shared 2025 test set. Where reliable RFC822 headers and recipient/URL features

are available, typed entity relations are a promising research direction, particularly for benign-body padding. Nevertheless, the 2026 experiment does not yet isolate a generalisable benefit from entity relations, and its apparent advantage should remain provisional.

5.10 Integrated comparison of the three studies

Tables 5.8 and 5.9 place the three methods alongside one another. They distinguish the original publications from the student experiments and make the different prediction targets and input requirements visible. This distinction is essential: results from separate datasets are useful for understanding each methodology, but they are not a controlled head-to-head leaderboard.

Study	Dataset and prediction target	Core methodology	Main strengths	Principal limitations
2024: Al-Subaiey et al. [1]	Merged Kaggle collection (Enron, Ling, CEAS, Nazario, Nigerian Fraud and SpamAssassin); mixed spam/ fraud/phishing vs legitimate labels. 82,068 cleaned student records.	Text cleaning and TF-IDF; linear SVC (with RF and MNB comparators); LIME local word explanations; original study also proposes a web interface.	Simple to train and deploy; transparent feature basis; LIME explanations; strong within-corpus scores.	Vocabulary and campaign dependence; mixed positive class; original website usability not tested in my run.
2025: Hosseinzadeh et al. [2]	Kaggle Phishing Emails: 18,650 original; 18,631 usable records; safe vs phishing text labels.	BERT contextual features with multi-head attention, CNN-1D and BiGRU; published hybrid uses MGO tuning. Compared plain BERT with fixed (non-MGO) hybrid.	Captures richer language context; directly tests whether added layers help beyond BERT.	Higher GPU cost; repeated texts not removed before split; run did not execute MGO.
2026: Shang et al. [3]	Raw RFC822 mail from Enron, Phishing Pot, Nazario and SpamAssassin; benign vs broader malicious mail, including spam; 28,386 assembled records.	40 typed entities; five relation functions; 1,600 matrix features; PSO feature mask; linear SVM; Good Word body-padding test.	Can use header and entity relationships; retains performance under studied text-padding perturbations.	Source/class confounding in my corpus; identical feature-ablation results; approximated lexicons; requires complete raw emails.

Table 5.8: Side-by-side comparison of methods, datasets, evidence and limitations

Study / selected model	Published F1	My F1	My test size	Meaning of comparison
2024 linear SVC / TF-IDF	≈0.9900	0.991438	16,414	Near the paper's reported result on a related merged-data protocol; not independent generalisation evidence.
2025 BERT hybrid (fixed run)	0.9643 (published MGO hybrid)	0.963970	1,864	Numerical similarity does not imply exact reproduction. My MGO was disabled.
2025 plain BERT (extra baseline)	Not a published standalone baseline in this study	0.970966	1,864	Direct within-my comparison: better than my fixed hybrid on the identical test examples.
2026 PhishSigma++ style	0.9675	0.999851	7,097	Near-perfect my value is strongly threatened by source/label coupling and is not comparable with the original paper.

Table 5.9: Published results versus student experimental results (F1-score)

In the 2025 paired experiment, standard BERT also required approximately 16.61 minutes of training versus 17.66 minutes for the fixed hybrid, with 12 versus 22 missed phishing emails (false negatives). For 2024, the best within-study method was TF-IDF plus linear SVC (F1 = 0.991438). In 2026, entity counts only, dense relations and PSO-selected relations all produced the same displayed F1 = 0.999851, so the supplied run did not demonstrate a separate gain from PSO or relation selection.

5.11 Which study is best? A criteria-based decision

A single winner cannot be established from the three headline F1-scores. The experiments use different datasets, label meanings, split sizes and feature access: 2024 and 2025 classify primarily email text, whereas 2026 uses full RFC822 structure. In addition, the 2025 duplicate issue and the 2026 source-label correlation can artificially make within-corpus results look stronger. Consequently, the valid judgement is a conditional recommendation based on the evidence available, not a universal performance ranking (Table 5.10).

Decision criterion / setting	Preferred approach under current evidence	Reason	Qualification
Practical, interpretable and relatively lightweight email-text filtering	2024 TF-IDF + linear SVC	F1 = 0.991438 in its experiment; LIME explanations and classical ML deployment are straightforward.	Best practical provisional recommendation; evaluate on a new phishing-only, source-disjoint dataset before deployment.
Context-sensitive email-text detection where a GPU is available	2025 plain fine-tuned BERT	Beat the 2025 fixed hybrid on the same 1,864-message test set (F1 0.970966 vs 0.963970), with fewer false negatives.	The 2025 published MGO-optimised model was not fully reproduced; group duplicate texts before rerunning.
Raw-email structural indicators and Good Word padding resilience	2026 PhishSigma++-style typed relations (research candidate only)	Combines headers/domains and typed relations; maintained performance against the student padding perturbation.	Not yet an evidence-based overall winner: source bias and identical ablation scores need independent testing.

Table 5.10: Evidence-based selection of an existing detection approach

Overall selection for this dissertation. The 2024 TF-IDF + linear SVC approach is the most defensible provisional choice if the priority is an efficient, explainable phishing-email text detector backed by the currently available evidence. This recommendation is conditional, not a finding of universal superiority. Within the 2025 experiment specifically, plain BERT is the better tested model; the additional unoptimised hybrid layers did not justify their complexity. The 2026 typed-relation approach is potentially valuable for structure-aware robustness, but the unusually high F1 is not reliable evidence of superiority until source-held-out tests and corrected ablations are completed.

Minimum validation required before a definitive three-way ranking: use one common raw-email benchmark with consistent phishing-versus-legitimate labels; derive a text-only view for TF-IDF and BERT; separate records by source/campaign and duplicate group; repeat test seeds; compare F1, phishing recall, false positives, explanations, robustness and inference cost; and report which method has access to extra header features. Without that shared protocol, a "best across 2024–2026" claim would be misleading.

Chapter 6

Conclusion and Future Work

6.1 Summary of objectives and findings

This dissertation has examined three existing email detection methodologies rather than designing a new framework. The 2024 controlled reproduction shows TF-IDF + linear SVC outperforming Random Forest and Naive Bayes on its merged dataset, with F1 = 0.9914. The 2025 experiment shows standard BERT outperforming the fixed BERT/attention/CNN/BiGRU hybrid on the same test set, with F1 of 0.9710 against 0.9640. The 2026 reproduction produces an unusually high F1 = 0.9999 for a typed-relation classifier and greater robustness than a token baseline on its constructed corpus, but severe source-label coupling and uninformative ablations preclude a confident claim about real-world deployment effectiveness. These results answer the project's research question by demonstrating that practical selection depends on evaluation design, dataset provenance, interpretability, robustness and operational cost, not only on the largest reported accuracy figure.

6.2 Implications

The findings show why source studies must be read beyond their headline metrics. Paper-inspired reimplementations are useful for testing conceptual claims, but precise experimental equivalence depends on original train/test assignments, detailed preprocessing, architecture parameters and source code. A simpler contextual baseline can be stronger than a more elaborate fixed hybrid. Likewise, exceptionally high scores can be a warning of convenient shortcuts in the dataset rather than evidence of superior phishing intent recognition. Transparent uncertainty and negative results are meaningful contributions to comparative computing research.

6.3 Future work: validation priorities

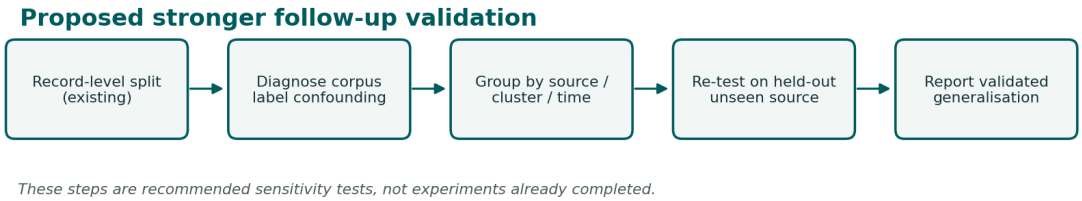


Figure 6.1: Recommended source- and duplicate-disjoint evaluation extension.

Priority one is to repeat the 2025 experiment after excluding or grouping duplicate email texts before splitting; this will quantify any optimistic bias in both BERT and hybrid results. Priority two is to rebuild the 2026 raw-email corpus with verified SpamAssassin and Nazario parsing and more balanced class provenance, then use leave-one-corpus-out and campaign/time-disjoint tests (Figure 6.1). Priority three is to repeat the 2026 feature ablations with feature checks to understand why all variants obtained identical metrics. Priority four is to evaluate computational overhead and inference latency consistently. A shared raw-email dataset providing both text and structured headers would make a direct three-method ranking possible in a later study.

6.4 Final statement

This research therefore identifies a conditional practical preference rather than an absolute winner. On the supplied evidence, the 2024 linear SVC approach offers the clearest balance of strong in-corpus performance, explainability and relatively simple deployment. For text requiring contextual modelling, plain BERT is supported by the direct 2025 paired comparison, outperforming the tested fixed hybrid. The 2026 PhishSigma++-style method demonstrates the potential of entity relations and padding resistance, but its apparent near-perfect result cannot be trusted as a generalisation claim while source-label confounding remains unresolved. The central contribution is a transparent, methodologically cautious comparison that tells practitioners what can—and cannot—be concluded from these three studies.

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Appendix A

Reproduction Artefacts and Links

A.1 Project source code repository

All notebooks, source code and generated outputs for this thesis are available in the project GitHub repository [4]:

GitHub: `https://github.com/Azeem74812/Msc-Thesis-Project.git`

A fully reproducible archival package should additionally contain the original `.ipynb` files, environment requirements, downloaded dataset manifests and SHA-256 checksums, executable code, saved train/test assignments and all generated figures.

Case	Uploaded evidence	Datasets	Main generated outputs
2024	Phishing_2024_vs_Baseline_Comparison.ipynb (Colab)	Kaggle merged dataset	Three confusion matrices; `comparison_ours`; LIME; `paper_vs_ours`
2025	Phishing_2025_vs_BERT_Complete_Comparison.ipynb (Colab)	Kaggle phishingemails	BERT/hybrid metrics; confusion matrices; McNemar
2026	Phishing_2026_PhishSigma_vs_Baseline_Comparison.ipynb (Colab)	Enron / Phishing Pot / Nazario / SpamAssassin	PSO history; ablations; Good Word curve; `comparison_ours`

Table A.1: Experimental evidence traceability

A.2 Main source URLs

- Project repository: https://github.com/Azeem74812/Msc-Thesis-Project.git
- 2024 paper: https://doi.org/10.1016/j.compeleceng.2024.109625
- 2024 dataset: https://www.kaggle.com/datasets/naserabdullahalam/phishing-email-dataset
- 2025 paper: https://www.nature.com/articles/s41598-025-20668-5
- 2025 dataset: https://www.kaggle.com/datasets/subhajournal/phishingemails
- 2026 paper: https://arxiv.org/abs/2605.11619
- Enron: https://www.cs.cmu.edu/~enron/
- Phishing Pot: https://github.com/rf-peixoto/phishing_pot
- Nazario: https://monkey.org/~jose/phishing/
- SpamAssassin: https://spamassassin.apache.org/old/publiccorpus/

A.3 Reproduction environment and run notes

2024: Google Colab CPU and Python scientific stack. The notebook downloads the merged Kaggle CSV, trains TF-IDF / MNB, Random Forest and SVC and saves test predictions and plots. **2025:** Google Colab T4 GPU; `bert-base-uncased`; 256-token inputs; batch and optimiser controls as listed in Chapter 4. **2026:** Google Colab CPU; raw RFC822 email parsing; typed feature construction; PSO and linear SVM. The 2026 feature extraction printed a runtime of approximately 123.9 minutes and the PSO search 18.7 minutes on the 28,386-message assembled corpus.

Appendix B

Additional Published Results and Interpretive Notes

B.1 2025 regularisation and optimiser evidence

Configuration	Accuracy	Precision	Recall	F1
Without L2	0.9629	0.9398	0.9555	0.9476
Without dropout	0.9595	0.9346	0.9554	0.9449
With L2 and dropout	0.9722	0.9426	0.9871	0.9643

Table B.1: Published 2025 ablation values (original study)

Optimiser	Accuracy	F1
GWO	0.9443	0.9289
WOA	0.9409	0.9251
SSA	0.9426	0.9302
AVOA	0.951	0.9368
GA	0.9392	0.9258
PSO	0.9477	0.9366
PUMA	0.9544	0.9408
MGO	0.9722	0.9643

Table B.2: Published 2025 optimiser results (original study)

These tables are drawn from the originating study as transcribed in the earlier project guide and are presented for literature completeness. They are not outcomes produced in the current uploaded student 2025 run. The original article reports inconsistent headline and detailed values; readers should refer to the primary publication when quoting a specific experimental configuration.

B.2 2026 published baselines

Method	Precision	Recall	F1
RuleIOC	0.683	0.1193	0.2031
SpamBayes	0.9967	0.9227	0.9583
SpamAssassin	0.8739	0.087	0.1583
DistilBERT comparator	0.9356	0.9379	0.9367
BERT comparator	0.9565	0.7786	0.8584
E-PhishGen / Naive Bayes	0.9793	0.784	0.8692
PhishSigma++	0.9736	0.9614	0.9675

Table B.3: Published 2026 clean-data literature comparison